



BUILDING THE TARGET

Paul Speaker

Target Construction

- Objective of data science is usually to build predictive models for a target variable
- Often, the target variable does not exist in the raw data
 - Real life is not like Kaggle problems!
 - Creating the target variable can sometime be more difficult than building the predictive model
- Many (most?) of the challenges are caused by working with data across time
- Feature engineering the X's across a dataset is also a challenge
 - These challenges often relate to the dimension of time

More on this Tuesday!!

Mechanics of Target Generation

- Suppose customer data is stored like below
- We want to build an churn model for customers
- “Everybody dies” model is boring, so we work on an annual model
- Data is in the wrong format

- Customer 1 was customer for 11 years → 11 rows in modeling dataset

- Target for 11 rows: 1 yes, 10 no's Left in 2017 (2007-2016, X = no)

- 2 data prep needs

- Get all of those extra rows

- Fields:

- CustomerID

- Year

- Year.Born and Male.Female

- Build the target (new column)

Target is “did they churn?”

E.g. #1: 2007 no, 2017 yes

Notice - data set unbalanced
(many more no than yes)
unavoidable in this case!

	Customer	First_Year	Last_Year	Year.Born	Male.Female
1	1	2007	2017	1961	F
2	2	2007	2016	1953	F
3	3	2011	2020	1957	F
4	4	2006	2014	1988	M
5	5	2005	2018	1965	F
6	6	2009	2019	1979	F

key is customer + year combination (can keep separate cols)

Example: Customer churn prediction

- (One) strategy of adding rows
 - Catalog all years
 - Every year from earliest First_Year to last Last Year
 - Can manually define year data frame with a sequence of those years
 - Do a Cartesian join between year dataframe and raw dataset
 - Filter where year is between First Year and Last Year (inclusive)
 - Note that these last 2 steps can be done in a single SQL statement

Cartesian - join everything to everything

	Customer	First_Year	Last_Year	Year.Born	Male.Female
1	1	2007	2017	1961	F
2	2	2007	2016	1953	F
3	3	2011	2020	1957	F
4	4	2006	2014	1988	M
5	5	2005	2018	1965	F
6	6	2009	2019	1979	F

Example: Customer churn prediction

- Now we have a dataset with all of the row, but we need to have a target
- Idea for target: we need to know, for each year, we need to “look ahead” to see if there is a row next year—joining across years
- (A) strategy for building the target
 - Create a “Next Year” field which is Year + 1
 - Use a (One-sided) Join of the dataset to itself where key is Next Year = year
 - If there is a match, no churn. If missing, there is churn

	Customer	First_Year	Last_Year	Year_Born	Male_Female	year
1	1	2007	2017	1961	F	2007
2	1	2007	2017	1961	F	2008
3	1	2007	2017	1961	F	2009
4	1	2007	2017	1961	F	2010
5	1	2007	2017	1961	F	2011
6	1	2007	2017	1961	F	2012

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Example: Order prediction

- Another problem: predicting whether customers are going to order products in the next month *target now in units of month, not year (monthly churn)*
- Raw dataset is order history (headers below)
 - Raw data contains history of orders
 - These are the “yes” targets
 - Dataset doesn't have orders not made
 - Need to get those “no”'s into the dataset

	CustomerID	Product_ID	Year	Month	Quantity
1	386	9	2010	5	93
2	874	6	2015	9	10
3	373	7	2010	6	17
4	943	3	2010	5	52
5	850	9	2015	5	36
6	696	4	2016	1	38

Best Practices of Building a Good Target

easily understood BY OTHERS - nontechnical?

- The target of a predictive model should
 - **Be easily understood and constructible from the data available** need target! if not constructible, project over!
 - **Be tied to the decision the model is meant to support**
 - **Deviations should carry uniform loss** more on this later
 - **Should not contain information leakage** more on this later
 - **Be built with the right prediction period and horizon** more on this later

time, time, time!!!!
- This is harder than it sounds!

The Nature of Time in Data Science

- The fundamental thing to consider about time in data science is that **time is two-dimensional**
 - **Event time**: the time at which an event happens
 - **Prediction/Observation time**. The time at which a prediction or observation is made
 - For predicting something in December 2026, there is only one event time, but we can predict from many horizons out
 - We can also make observations of historical data, and that can change over time
 - More on this next class!
 - To build a data science model, we have to use information available at prediction time
 - The prediction time is tied to when a decision needs to be made
 - **Target leakage**. Using information only available after the prediction time
- mistake in a model - using data that won't be available (e.g., use last month to predict this month)**

**$R^2 = 1$, using
volume (kg) to
predict volume
(lbs)**

Data Transformation and Loss Functions

- Even when you have identified the right target, it might make sense to do a data transformation to get the most impactful model
- Don't want to treat equal loss values unequally
- Don't want to treat unequal loss values equally
- Examples:
 - Pizza delivery, Inventory predictions pizza - ok to deliver early, late is a problem! (skew loss function to the later side) (20 min early = 5 min late)
 - Asymmetric loss when skewing to one side better than alternative
 - Mortgage payoff predictions
 - Log transform
 - Delivery delays
 - Binning
 - Customer churn
 - In-class assignment!
- Later: what does this tell us about loss function to minimize?

Target Examples

- Let's think about some examples of target building
 - Collections: you want to build a model that prioritizes accounts for collection actions.
 - Customer Churn. Which customers should be focus on?
 - Delivery problems. What is the target?